



AM Management Group Website v3

Installation, Operation & Validation Manual

Prepared by Iconic Soft Ltd. · Version 3.0.0 · 18 September 2026
Next.js 16 · Express 5 · Prisma 7 · PostgreSQL 16

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Client	AM Management Group Sdn. Bhd.
Prepared by	Iconic Soft Ltd.
Version	3.0.0
Date	18 September 2026 (v3)
Stack	Next.js 16 · React 19 · Tailwind CSS 4 · Express 5 · Prisma 7 · PostgreSQL 16

1. Overview

Version 2.0 does two things:

1. **Redesign.** The public website gets a premium corporate theme with a compact hero, an animated group-structure infographic, crossing "safety tape" ticker banners, an animated figures infographic and a thin banner on every inner page. The old image slider is removed. All existing website text is kept.
2. **Database move.** The API moves from MySQL to PostgreSQL. Endpoints, request and response formats are unchanged, so the admin dashboard and website keep working as before.

Package contents

```
am-management/  
├── frontend/           Next.js website + admin dashboard  
├── backend/           Express REST API (/api/v1/*)  
│   ├── prisma/schema.prisma  
│   ├── prisma/migrations/20260918000000_postgresql_init/  
│   ├── prisma/seed.ts  
│   └── scripts/migrate-mysql-to-postgres.ts  
├── docs/  
│   ├── MANUAL.md / MANUAL.pdf           this document  
│   └── am-management-v3-preview.html     single-file design preview  
├── docker-compose.yml           local PostgreSQL  
├── DESIGN-SYSTEM.md           colours, type, components  
└── README.md
```

2. Requirements

Item	Version
Node.js	20.19 or newer (22 LTS recommended)
npm	10+
PostgreSQL	14+ (16 recommended) — local via Docker, or managed (Neon, Supabase, Railway, AWS RDS)
Docker	optional, only for local PostgreSQL
Cloudinary account	existing, for image uploads
SMTP account	existing, for password-reset e-mails

3. Local installation

3.1 Database

```
docker compose up -d db
```

This starts PostgreSQL on `localhost:5432` with database `am_management`, user `am_user`, password `am_secret`.

3.2 API (backend)

```
cd backend
cp .env.example .env
```

Edit `.env`. The important values:

Variable	Meaning
<code>DATABASE_URL</code>	<code>postgresql://am_user:am_secret@localhost:5432/am_management?schema=public</code>
<code>DATABASE_SSL</code>	<code>false</code> locally, <code>true</code> for managed providers
<code>JWT_ACCESS_SECRETE</code> , <code>JWT_REFRESH_SECRETE</code>	long random strings (keep the existing spelling)
<code>APP_URL</code>	website URL, used for CORS and e-mail links
<code>SEED_ADMIN_EMAIL</code> , <code>SEED_ADMIN_PASSWORD</code>	first admin account

Variable	Meaning
CLOUDINARY_*, SMTP_*	copy from the current production settings

Then:

```
npm install          # also runs "prisma generate"
npm run db:deploy    # creates all tables
npm run db:seed      # admin user, 6 group companies, base settings
npm run dev          # API on http://localhost:5000
```

Check: opening `http://localhost:5000` shows "AM Management Group Backend Server IS Running..."

3.3 Website (frontend)

```
cd ../frontend
cp .env.example .env.local    # NEXT_PUBLIC_API_BASE_URL=http://localhost:5000/api/v1
npm install
npm run dev                  # http://localhost:3000
```

3.4 Command reference (backend)

Command	Purpose
<code>npm run dev</code>	development server with reload
<code>npm run build</code> / <code>npm start</code>	production build / run
<code>npm run db:deploy</code>	apply migrations (use in production)
<code>npm run db:migrate</code>	create a new migration after editing <code>schema.prisma</code> (development only)
<code>npm run db:seed</code>	insert baseline data (safe to repeat)
<code>npm run db:studio</code>	browse the database in a browser
<code>npm run migrate:mysql-to-pg</code>	one-off copy of live MySQL data

4. Moving live data from MySQL to PostgreSQL

Do this once, on a copy first, then for real during a short maintenance window.

1. Create the production PostgreSQL database and set `DATABASE_URL` (and `DATABASE_SSL=true`) in `backend/.env`.
2. Run `npm run db:deploy` to create the empty schema. **Do not run the seed** before importing.

3. Set `MYSQL_URL` to the current MySQL database, for example `mysql://user:pass@host:port/dbname?ssl={"rejectUnauthorized":false}`
4. Put the website in maintenance or stop admin edits.
5. Run `npm run migrate:mysql-to-pg`. All 13 tables are copied in dependency order inside one transaction. If anything fails, nothing is written.
6. Compare row counts (the script prints them) with MySQL.
7. Point the live API at PostgreSQL and redeploy.
8. Check: admin login, one company page, one project page with images, submit a test inquiry.

To repeat from scratch: `npm run migrate:mysql-to-pg -- --truncate`.

Images stay on Cloudinary; their URLs are copied unchanged.

5. Production deployment

Part	Where	Settings
PostgreSQL	Neon / Supabase / Railway / RDS	enable daily backups
API	Railway, Render or a VPS	<code>build npm run build</code> , <code>release npm run db:deploy</code> , <code>start npm start</code>
API on Vercel (optional)	<code>vercel.json</code> kept	set the same env vars
Website	Vercel	<code>NEXT_PUBLIC_API_BASE_URL=https://<api-domain>/api/v1</code>

After deploying:

- Add the website domain to `APP_URL` on the API (CORS).
- Set `NODE_ENV=production`.
- **Rotate all credentials** that were in the old project's `.env` files (database, JWT secrets, Cloudinary, SMTP). They were included in the previously shared ZIP.

6. Content management (admin dashboard)

Sign in at `/login` and open `/dashboard`. Administrators see every section below; normal users see only Overview and Profile.

Section	What it controls on the website
Overview	counts and recent activity

Section	What it controls on the website
Companies	group companies: name, category, logo, cover image, description, status. Mark one as main company to show it as the large navy card on the home page
Services	services listed on company pages and the Services page
Projects	project pages; tick featured to show on the home page (up to 6 small cards). Status (Completed / Ongoing) drives the home filter
Gallery	images for the Gallery page and company pages (the home page no longer has a gallery section)
News	news articles
Careers	job posts shown on the Careers page
Applications	CVs submitted from job pages
Inquiries	messages from the contact forms
Users	admin and user accounts
Profile	your own name, avatar and password
Settings	site-wide text, contact details, logo, hero images, social links, SEO

6.1 Settings keys

Key	Where it appears
<code>site.name</code>	header, footer, page titles
<code>site.tagline</code>	home hero headline and footer statement (e.g. <i>One Group. Multiple Businesses.</i>)
<code>site.description</code>	home hero paragraph, footer
<code>site.business_hours</code>	footer
<code>site.website_logo</code>	logo everywhere (falls back to the built-in logo)

The old `site.hero_banner_desktop` / `site.hero_banner_mobile` keys are no longer used (the image slider was removed) and their fields were taken out of the Settings screen. Existing values in the database are harmless. | `contact.phone` , `contact.whatsapp` , `contact.email` , `contact.address` | top bar, footer, contact page | | `social.facebook` , `social.linkedin` , `social.instagram` , `social.twitter` | top bar and footer icons (hidden when empty) | | `seo.title` , `seo.description` | search engine title and description |

6.2 Fixed figures

The figures 10+ years, 6+ companies, RM 2.66M documented contract value, RM 500K share capital, CIDB Grade G5 and 2012 are part of the design, not the CMS. To change them edit:

- `frontend/src/components/home/Statistics/Statistics.tsx`
- `frontend/src/components/home/Banner/Banner.tsx` (hero figures and tape texts)

7. Design summary

Full details are in `DESIGN-SYSTEM.md`.

Element	Description
Colours	navy <code>#234279</code> , harbour navy <code>#0F2447</code> , signal orange <code>#FB731F</code> , steel <code>#EEF2F7</code>
Typeface	Sora for headings (hero title 1.75rem → 2.5rem, section titles 1.5rem → 2.1rem, weight 600); Archivo for body
Motif	blueprint grid surfaces and orange/navy safety tape
Home hero	compact, with an animated diagram: the holding company at the centre, six compact, semi-transparent 3D glass orbs with large icons, thick gradient connectors with flowing light and travelling pulses, two counter-rotating orbit rings
Tape banners	two premium ribbons crossing at a slight angle — orange gradient (sectors) and navy with orange edges (credentials); fine inner rules, slow light sheen, tracked small caps, star separators; pause on hover
Spacing	compact scale: sections 48 px (mobile) → 64 px (desktop) vertical padding
Companies grid	a "Partner with the group" card fills the last row when the company count leaves a gap
Featured projects	grid of up to 6 small cards (3 per row on desktop) with All / Completed / Ongoing filter
Figures	count-up numbers with pictograms (1 block = RM 100K)
Inner pages	thin blueprint band with breadcrumb and running tape edge
Accessibility	keyboard focus ring, reduced-motion support, labelled controls

To review the look without installing anything, open `docs/am-management-v3-preview.html` in a browser. It is a static copy of the home page and an inner-page banner, with sample project and gallery photos.

8. Testing & validation

8.1 Checks performed on this package

#	Check	Result
1	Syntax parse of all 214 TypeScript/TSX files (frontend + backend + seed + scripts)	Pass — 0 errors

#	Check	Result
2	Resolution of every local import (316 frontend, 171 backend)	Pass — only <code>generated/prisma</code> is unresolved, which is created by <code>prisma generate</code> during install
3	Prisma schema vs PostgreSQL migration: 13 tables, every column, nullability, enum types, defaults, unique keys, indexes, foreign keys and delete rules	Pass — 0 mismatches
4	Case-insensitive search on PostgreSQL: <code>mode: "insensitive"</code> added to 48 search filters in 10 services; all sit inside typed <code>Prisma.*WhereInput</code> objects	Pass
5	Client/server component boundaries: every file using React hooks or event handlers has <code>"use client"</code>	Pass
6	Custom theme utilities used by components exist in <code>globals.css</code>	Pass
7	Secret scan of the package (<code>.env</code> , passwords, Aiven certificate)	Pass — only <code>.env.example</code> files included
8	Preview rendered in Chromium at 1440 px and 390 px: horizontal overflow, script errors, visual review	Pass after fixes below
9	Dead-code scan: every component/module is imported somewhere; every asset is referenced; every npm dependency is used (except <code>react-dom</code> , required by Next.js)	Pass after cleanup
10	Featured projects filter (All / Completed / Ongoing) in the preview	Pass
11	HTML structure: one <code><main></code> per page, one <code><h1></code> per page	Pass after fixes below
12	Reduced-motion mode renders without script errors	Pass
13	Motion test on 13 pages: nothing left hidden after scrolling (normal and fast scroll), count-up 0 → target, hover lift, reduced motion shows everything static	Pass
14	Functional test: 23 routes, dropdown, mobile menu, filters, lightbox, thumbnails, form validation, pre-filled enquiry	Pass

8.2 Issues found and fixed during testing

Issue	Fix
The two crossing tape banners overlapped so much that half of the orange sector tape was hidden	Reduced the angles (-1.2° / 0.9°), put the orange tape on top, adjusted spacing
Font fallback on systems without the web font showed a serif face in some paragraphs	Added Helvetica / Arial to the font stack

Issue	Fix
Hero infographic labels on the right-hand nodes could run past the frame on narrow screens	Labels now align towards the centre
Dead import of a removed component in the home page	Removed
Hero title too large and heavy	Changed to Sora 600, 2rem → 3.15rem
Image slider under the tapes (and its Settings fields) no longer wanted	Removed component, images, Settings fields
Large featured-project tile	Removed; small-card grid only
Home "Our Work In Pictures" section no longer wanted	Removed component and its data request
13 pages nested a second <code><main></code> inside the layout's <code><main></code> (invalid HTML, confuses screen readers)	Inner wrappers changed to <code><div></code>
Job detail page showed the job title twice as <code><h1></code> (banner + body)	Body title changed to <code><h2></code>
Home page fetched companies and gallery on the server but never used them	Removed; only featured projects are fetched
Company grid left an empty slot with 5 companies	Added "Partner with the group" card to fill the row
Headings too large, too much empty space	All <code>h1</code> / <code>h2</code> two steps smaller site-wide; section, banner and hero padding reduced about 40%
Unused files: 11 images, 5 default SVGs, 4 static data files, Redux store, 3 unused components, 2 hooks, 1 modal, API endpoint map, agent notes	Deleted
Unused npm packages: swiper, framer-motion, sweetalert2, react-hot-toast, @reduxjs/toolkit, react-redux, nodemailer, @types/nodemailer, shadcn-ui	Removed from <code>package.json</code> ; stale <code>package-lock.json</code> removed (regenerated by <code>npm install</code>)

8.3 Not tested here (network was unavailable)

`npm install`, `next build`, the TypeScript type check and a live PostgreSQL run could not be executed in the build environment. Run the acceptance checklist below once after installation; allow about half a day for any build fixes.

8.4 Acceptance checklist

Build

- ☐ `backend`: `npm install` and `npm run build` finish without errors
- ☐ `frontend`: `npm install` and `npm run build` finish without errors
- ☐ `npm run lint` in `frontend` shows no errors

Database

- ☐ `npm run db:deploy` creates 13 tables
- ☐ `npm run db:seed` prints admin, 6 companies, settings
- ☐ (live data) `migrate:mysql-to-pg` row counts match MySQL

Website

- ☐ Home: hero diagram animates, tapes scroll, "Who we are" follows directly after the tapes
- ☐ Home: figures count up when scrolled into view
- ☐ Home: companies, featured projects (All / Completed / Ongoing), gallery load from the API
- ☐ Every inner page shows the new banner with breadcrumb
- ☐ Company page shows its cover image in the banner
- ☐ Header dropdown lists companies; mobile menu opens and closes
- ☐ Contact form and job application submit successfully
- ☐ Layout checked at 390 px, 768 px, 1280 px and 1920 px
- ☐ With "reduce motion" enabled in the operating system, animations stop

Admin

- ☐ Login, logout, password reset e-mail
- ☐ Search in Companies / Projects / Users finds results regardless of upper/lower case
- ☐ Create, edit and delete one record in each section
- ☐ Image upload to Cloudinary works
- ☐ Changing `site.tagline` or `site.description` in Settings updates the home hero

9. Troubleshooting

Symptom	Cause / fix
Cannot find module '../..../generated/prisma/client'	run <code>npm install</code> or <code>npx prisma generate</code> in backend
P1001 Can't reach database server	check <code>DATABASE_URL</code> , that PostgreSQL is running, and <code>DATABASE_SSL</code>
self-signed certificate / SSL error	set <code>DATABASE_SSL=true</code> for managed databases
Website shows default text and no companies	<code>NEXT_PUBLIC_API_BASE_URL</code> wrong, or API not running
Not allowed by CORS	add the website URL to <code>APP_URL</code> on the API
Remote images do not appear	image host must be reachable over https; <code>next.config.ts</code> already allows all hosts
Admin search misses results	confirm the backend is v2.0 (search uses <code>mode: "insensitive"</code>)

Symptom	Cause / fix
Migration script: duplicate key	target not empty — rerun with <code>-- --truncate</code>

11. What's new in v3

Area	Change
Content	Company profile 2026 used as the source: company story, CIDB G5 (B04, CE21, M15, valid to Jun 2027), registration and incorporation details, leadership, 7 projects with sites, dates and contract values
Figures	Home figure now RM 2.66M documented contract value (two programmes) instead of RM 4.8M+, shown as an animated bar chart
Charts	Projects page: animated status donut, projects-by-state bars and contract timeline, computed live from published projects
Photos	Deliberately few: 9 real photos in <code>frontend/public/images</code> (8 project, 1 head office). Duplicates and stock images from the PDF excluded
Contact	Real address and phones; both emails shown everywhere: info@ammanagement.com.my (primary) and ammanagement2012@gmail.com (new Settings key <code>contact.email_alt</code>)
Companies	Hidensypro Sdn. Bhd. (machinery), BM Magnitude Services (cleaning), CM Plantation Services (plantation), AM Multi Trade Empire (mini market), MA Travel and Tour Sdn. Bhd. (ticketing)
Seed	<code>npm run db:seed</code> loads settings, 6 companies, 18 services, 7 projects, 6 gallery photos, 3 news items and 4 jobs from <code>prisma/seed-data.json</code>
Banners	Inner-page banners are calm: no moving stripes, solid orange rule. Tape stripes outside the hero are static
Excluded	IC numbers, bank statements, EPF/SOCSO records, shareholdings and personal ages from the PDF are not used anywhere

Bugs fixed in v3

Bug	Fix
API and dashboard rejected site-hosted image paths (<code>/images/...</code>), so seeded photos could not be edited	URL validators accept absolute URLs or <code>/path</code>
Seed script used <code>__dirname</code> , which crashes in an ES-module backend	Uses <code>import.meta.url</code>
Seed used a non-existent <code>prisma.gallery</code> model	Uses <code>prisma.galleryImage</code>
Client settings hook kept its own stale defaults (phone "+1 (234) 567-890", old address)	Hook now reuses the single settings module

Bug	Fix
Settings mapper crashed when no hero-banner key existed (<code>undefined.split</code>)	Guarded

Motion & animation (v3.1)

Change	Detail
One motion language	Single easing curve <code>cubic-bezier(.22,1,.36,1)</code> and three durations (200 / 420 / 750 ms) used site-wide (<code>--am-ease</code> , <code>--am-dur-*</code>)
Scroll reveal	Section headings, cards and grids fade up once with a 70 ms stagger (<code>ScrollReveal</code> , <code>data-reveal</code> , <code>data-reveal-stagger</code>). Content is never hidden without JavaScript or with reduced motion; a scroll sweep reveals anything skipped by fast scrolling
Calmer hero	Hub ring 6 s → 14 s, light flow slower and sparser, pulses every 3.6 s, orbs float 3 px instead of 5 px, softer entrance with a light blur-in
Card hover	4 px lift on desktop pointers only, no delay after the entrance
Charts	Home contract-value bars, Projects donut / state bars / timeline animate once in view

Motion bugs fixed

Bug	Fix
Count-up numbers showed the final value, then jumped back to 0 when scrolled into view	Reset to 0 on mount; server still renders the final value
A glowing dot sat at the top-left of the hero diagram for the first 2 s	Pulse starts invisible until its animation begins
Stagger delay also delayed hover effects on cards	Delay removed once the entrance finishes

Please keep in mind

- Five appendix projects (Melekek, Sungai Petai, Kota Syahbandar, Bahau, Paya Rumput) have no status in the PDF; they are set to Completed.
- The Melaka housing and shop-office programme stays Ongoing although its end date (31 Jan 2026) has passed, as stated in the PDF.
- The CIDB certificate shows status "DORMAN"; the site states the registration and grade only.

10. Support

Iconic Soft Ltd. — *Your Vision, Our Code.* <https://iconicsoftltd.com>